PakEmis Sector Technical Note TN02

A Plant-Resolved, Fuel-Constrained Inventory of Greenhouse-Gas and Air-Pollutant Emissions from Electricity Generation in Pakistan

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Abstract

We reconstruct Pakistan's thermal generating fleet plant by plant beneath the national fuel statistics it is normally summarised by, and report electricity-generation emissions of 45.59 Mt CO_2e for FY2023–24: 40.04 Mt from grid generation (1A1a; the selected member of a structural family: 38.45 on a fuel-property alternative, 44.05 on a coverage alternative and 57.82 on the legacy published account) and 5.55 Mt from industrial captive gas self-generation (1A2), together 10.2 % of a ten-sector national inventory, on a single reference year.

The activity architecture is a control total with a ledger underneath it. The regulator's national fuel table—in tonnes of oil equivalent, already an energy unit—sets each fuel's magnitude; a ledger parsed from the same report's plant tables, covering 99.0 % of thermal generation and validated per fuel against a generation table the parser never reads, sets the structure. The ledger turns the fleet's one unpublished composition question, the imported-bituminous/Thar-lignite blend, from an inference into an observation: mine-mouth plants are 75.2 % of coal generation, and the lignite share is 0.79 with its one unresolved plant carried as bounds (0.752 to 0.835). On an energy basis that blend moves the coal factor by 6.8 % end to end rather than the factor of two it would be worth per tonne.

Forcing the plants and the national table to reconcile also tests the statistics themselves, and two findings are published rather than absorbed. Plants metered in MMBTU—already energy, so no calorific assumption enters—imply 48.7 % net efficiency, consistent with the national table being on a net calorific basis, which had been this sector's largest declared risk. But the national coal energy implies 24.2 TJ/Gg over a fleet that is 92 % lignite by mass, and the report's own per-plant heat rates—an LHV channel with Table 3 nowhere in the loop—put the Thar plants at 11.1 TJ/Gg, squarely IPCC lignite: the official conversion is high by a factor of 2.01 on this channel, worth 1.6 Mt; and the plant sums bracket the national gas row only when the K-Electric system is excluded, while the furnace-oil sums sit closer to the national row with it included in four of five years, so whether the official table already contains K-Electric's fuel is not established. The first question is now decided and the second is not. The sector books *grid_stations_rank_default* 40.04 Mt, the stations' own metered coal at default rank calorific values, because that account implies plausible net fleet efficiencies in nine of nine panel years where the national conversion manages two; the conversion is retained as *grid_nepra_reference_legacy* 57.82 Mt, with *grid_plant_cv_restated* 38.45 Mt and *grid_plus_ke* 44.05 Mt beside them—competing definitions of the control total, not sensitivities around a settled one, and not of one kind: the restated case is a fuel-property alternative on the reference's own fleet, and the K-Electric case is a coverage alternative, priced beside the reference and not added to it, because adding it would count that fuel twice if the national statistic already contains it. The UNFCCC filing's 2 % solid-fuel agreement is NOT counter-evidence on the coal case, as an earlier edition of this abstract claimed: the filing is compiled from the same energy balance the national conversion reports, so it agrees with that conversion by provenance. The parametric interval below is conditional on the selected member.

Captive gas self-generation rests on a register of 1,105 plants, assembled from related administrative records, that sums to 288.01 MMCFD against the regulator's published 288; it is one fuel in one use, a floor on industrial combustion, not an estimate of it. Air pollutants are published as an explicitly uncontrolled upper bound—an uncontrolled screening scenario of 209 kt SO_2, 113 kt NO_x, 362 kt filterable particulate of which 34 kt is $PM_{2.5}$, each coal rank on its own AP-42 section since the Thar-lignite split—because no dataset records which controls are fitted where, and applying an unpublished removal efficiency would invent the number that matters most. The gas NO_x factor blends the boiler and turbine chapters by the ledger's heat-rate classification of the fleet, 80 % combined-cycle, with the unresolved share reported rather than assigned.

Keywords: electricity generation, captive power, energy-basis activity data, coal blend, industrial combustion, uncontrolled emission bound, AP-42, IPCC 2006 Guidelines, Pakistan, MRV

PakEmis Sector Technical Note TN02. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

Release provenance. This report documents one release of PakEmis and is generated against it. Where a sector report and the release's generated product catalogue (`outputs/pakemis_product_catalog.csv`) disagree, the catalogue is authoritative: sector reports are revised on their own cadence and may lag the products.

Release version	`1.0.0rc67`
Commit	`1876f398f85c`
Reporting period	FY2023-24 (2023-07-01 to 2024-06-30)
Baseline configuration	S (survey basis)
National total	444.96 Mt CO_2e (AR5) under S, the reference; 450.20 Mt under L
Release manifest	`13462fb4c2d1ea62` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

Pakistan's thermal generating fleet is a small, fully licensed, fully metered population: roughly seventy plant–fuel series across three systems—the CPPA-G market operator's independent producers, four legacy public generating companies, and the vertically integrated K-Electric utility in Karachi—burning imported bituminous coal, domestic Thar lignite, pipeline gas, regasified LNG, furnace oil and diesel. Its regulator publishes both a national fuel account and, in the same annual report, plant-level fuel and generation tables that national inventories, including this one until now, have left unread.

This paper reconstructs that fleet plant by plant and forces it to reconcile with the national statistics above it. The architecture is a control total with a ledger underneath: the regulator's national fuel table—in tonnes of oil equivalent, already an energy unit—sets each fuel's magnitude, and a ledger parsed from the plant tables, covering 99.0 % of thermal generation, sets the structure beneath it: which plants, burning which coal, in which province, at what implied efficiency. The reconciliation runs both ways. The ledger answers questions the national table cannot—the coal blend becomes an observation rather than an inference—and it interrogates the national table itself, yielding two findings about Pakistan's official energy statistics that the aggregate figures alone could never expose.

On that basis we report 45.59 Mt CO_2e for FY2023–24 across two IPCC categories—grid generation (1A1a) at 40.04 Mt and industrial captive gas self-generation (1A2) at 5.55 Mt—on a single reference year, 10.2 % of a ten-sector national inventory, with an air-pollutant layer published as an explicitly uncontrolled bound. The grid figure is the selected member of a structural family (38.45 and 44.05 Mt are the plant-implied-calorific and K-Electric-inclusive readings, the first a fuel-property alternative and the second a coverage alternative kept apart from it), and no parametric interval in this paper is quoted without that family beside it.

1.1. Contributions

1. **A plant fuel ledger** parsed from the regulator's own tables, covering 99.0 % of FY2023–24 thermal generation, validated per fuel against a generation table the parser never reads and screened row-by-row against fuel-intensity physics (Section 3).
2. **A coal blend read from plant identity** rather than inferred: lignite share 0.79, with the one plant whose origin its name does not state carried as bounds (0.752 to 0.835). On the energy basis this activity is kept in, the blend moves the coal factor by 6.8 % end to end rather than the factor of two it is worth per tonne (Sections 4 and 5).
3. **A physical energy closure** between plant meters and the national table. Its assumption-free leg—plants metered in MMBTU—implies 48.7 % net efficiency, evidence that the national table is on the net calorific basis the IPCC factors require, which had been this sector's largest declared risk (Section 6).
4. **Two findings about the official statistics**, published rather than absorbed: the national coal energy implies a calorific value 1.86 times what a 92 %-lignite fleet supports, and the plant sums bracket the national gas row only when K-Electric is excluded—each a question worth megatonnes, posed precisely enough to be settled by a tariff filing or a scope note (Section 6).
5. **Captive gas self-generation**, from metered gas reconciling with a 1,105-plant register to within 0.01 MMCFD, reported as a floor on industrial combustion rather than an estimate of it; and **an air-pollutant layer as a declared uncontrolled bound**, with the parameters that would turn it into an estimate named (Section 8).

Method shared with the other nine sectors—the module contract, the accounting closures, the correlated-uncertainty treatment—is documented separately (Ahmad, 2026).

2. Related Work

2.1. Fuel-based and generation-based power inventories

Power-sector emissions can be built from fuel consumed or from electricity generated with an assumed heat rate. The IPCC 2006 Guidelines describe the fuel route as Tier 1 and treat generation-based estimates as a fallback where fuel data are unavailable (IPCC, 2006), because the heat rate is an efficiency assumption that varies by plant, by load and by year,

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and because it is precisely the quantity a compiler is least able to verify.

Pakistan's regulator publishes fuel directly, so the fallback is unnecessary here. This is worth stating because a workbook consolidating Pakistani plant-level data—which supplied the captive register used in this paper—computes its CO_2 as kilograms per megawatt-hour derived from IPCC factors through an assumed plant-efficiency range. That is the generation route, and its output is not used here for that reason, while its plant register is.

2.2. *Coal quality and why the unit matters*

The two coals in Pakistan's fleet are imported bituminous, burned at Sahiwal, Port Qasim and Hub, and domestic Thar lignite. Published proximate analyses of Thar coal report moisture near 46 %, ash around 6.4 % and calorific values near 3030 kcal/kg—roughly half that of imported steam coal.

That difference is the whole issue. Emission factors expressed per tonne carry it in full; factors expressed per terajoule do not, because the calorific value is exactly what the differing energy content describes. The methodological literature on coal inventories treats blend uncertainty as a major term, and it is—on a mass basis. Section 4 shows it is a 6.8 % term on an energy basis, which changes what a compiler should worry about.

2.3. *Captive gas self-generation and the industrial-combustion gap*

Industrial self-generation sits awkwardly in inventory practice. IPCC assigns it to 1A2 with manufacturing-industry factors, but the data that would support it are usually collected—if at all—by an electricity regulator whose remit stops at plants selling into the grid.

For Pakistan the consequence is visible in the international comparisons: EDGAR (Crippa et al., 2025) reports Pakistani industrial combustion near 44 Mt while most national compilations report nothing, and the difference between a regulator's scope and an inventory's scope is a substantial part of that. The activity data that do exist come from a different regulator entirely—the oil and gas authority, which reports gas delivered to captive power because it regulates the pipeline rather than the plant.

2.4. *Uncontrolled factors and control-device inventories*

US EPA's AP-42 compilation (US Environmental Protection Agency, 1995) publishes emission factors for coal, oil and gas combustion predominantly as *uncontrolled* values, with separate treatment of control efficiencies, because what leaves a furnace and what leaves a stack differ by up to two orders of magnitude for particulate and by most of the mass for sulphur.

Applying uncontrolled factors to a fleet with precipitators and desulphurisation, without saying so, is a well-known way to overstate a power sector's air pollutants by an order of magnitude. Section 8 takes the position that where the control inventory does not exist, the honest output is the uncontrolled bound explicitly labelled as one—not a plausible-looking estimate resting on an invented removal efficiency.

3. Study Area and Data

3.1. *The fleet, and the ledger that represents it*

Pakistan's thermal fleet is reconstructed here as a plant fuel ledger parsed from the regulator's own plant tables (State of Industry Report, Tables 18, 21 and 26): 297 rows of plant × year × fuel quantity × net generation, FY2019–20 to FY2023–24, across the three systems those tables separate—the public generating companies, the CPPA-G market's independent producers, and K-Electric. At the FY2023–24 anchor year the ledger covers 99.0 % of thermal generation, and it is validated against a table the parser never reads: summed per fuel, it reproduces the report's own generation-by-fuel table to within 2.5 % for every fuel, and coal to 0.2 GWh across eight plants. Every row is additionally screened against fuel-intensity physics, because in an irregular document a misread column produces impossible plants, not subtle errors.

Units are preserved as the document states them, and they are load-bearing: gas metered in MMBTU is already energy, so those rows support comparisons with no calorific assumption; gas metered in MMCFT is a volume needing a heating value nothing here pins; coal and oil are masses. The distinction decides what each part of the ledger can prove (Section 6).

3.2. *The control total: national fuel in tonnes of oil equivalent*

The activity datum for the 1A1a estimate itself is the same report's national fuel account (Table 3), in tonnes of oil equivalent—an energy unit—converted to terajoules by one documented constant. The ledger structures this total; it does not replace it.

Fuel	**TJ**	**Share (%)**
Coal	206 606	37.9
Gas	313 243	57.5
Furnace oil	23 072	4.2
Diesel oil	1 811	0.3
Total	**544 732**	**100.0**

Table 1: Fuel input to main-activity electricity generation, FY2023–24, on the booked coal account. Coal is the second fuel by energy here and the first by emissions, because its factor is higher: it is 37.9 % of the energy and 51.6 % of the category's CO_2. On the aggregate fuel control booked before the coal re-base coal was also the largest by energy; whether it passed gas in this year is therefore a property of the coal account, not an observation, and it is the reason the coal question carries the weight it does.

3.3. *Captive gas*

The activity datum for the 1A2 category is the gas delivered to captive power, published annually by the Oil and Gas Regulatory Authority. The series runs FY2016–24 and is retained in full, with its own quality flags:

3.4. *The captive plant register, and why it matters*

A register of 1,105 physical captive plants was assembled from three sources: the SNGPL captive fleet (480 plants, Punjab and Khyber Pakhtunkhwa gas network), the SSGC geocoded captive list (557, Sindh and Karachi), and the

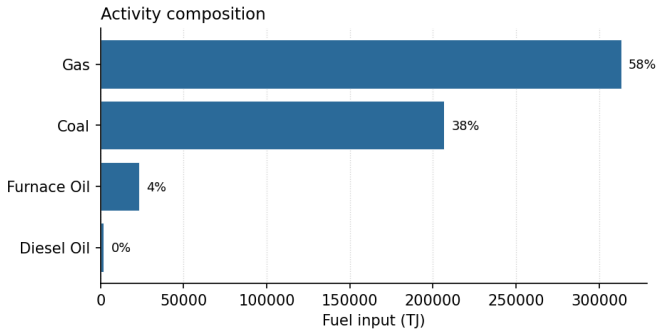


Figure 1: Activity composition before any emission factor is applied.

FY	MMCFD	Status
2016	389	verified
2017	393	verified
2018	380	verified
2019	321	verified
2020	290	verified
2021	203	provisional — read by OCR from a chart label in an image-scan edition
2022	243	verified
2023	123	verified; SSGCL restated upward in the following edition
2024	288	verified — the reported year

Table 2: Metered captive gas. The two flagged years are carried with their flags rather than smoothed, and neither is the reported year.

licensed generation-CPP register (83, national), deduplicated where a licensed plant is the same physical site as a gas connection.

Its metered gas sums to 288.01 MMCFD against OGRA's independently published national total of 288. Two sources built differently, from utility connection records and from a regulator's aggregate, agreeing to 0.01 MMCFD is the check that they describe the same fleet—and it is the reason this category is reported at all rather than left inside the gap.

The register also supplies a *measured* provincial split, only the second in this inventory after fugitive oil and gas: Sindh 166.3, Punjab 93.8, Khyber Pakhtunkhwa 13.3, Balochistan 1.8 MMCFD, with 0.6 in the Islamabad Capital Territory. The four province figures do not exhaust the register: 37 plants holding 12.3 MMCFD carry no province in the connection records, and that residual is redistributed pro rata over the assigned fleet rather than dropped—dropping it would apply shares summing to 0.957 and quietly lose four percent of the category (`data/province_shares.csv` declares the treatment). The fleet is dominated by textile plants (540 of 1,105).

3.5. Plant-wise generation, for the provincial proxy

Grid emissions are allocated by plant-wise thermal generation, from a consolidated series covering grid plants annually 2015–24 with province and district. Only thermal plants enter the normalisation: hydro, wind, solar and nuclear generate without combustion and must carry no share of a combustion sector's emissions. Appendix C treats what this replaced and why.

3.6. Plant-month generation, for the monthly layer

The national release resolves this sector's annual totals into fiscal months, and the evidence behind those months is this sector's to document. Two plant-month generation sources cover the panel, and which year is asked for decides which answers: FY2015–16 to FY2020–21 ride PBS's Electricity Generation Data (plant-by-plant, month-by-month), used wherever enough of a subitem's annual control sits in plants the publication actually labels; FY2021–22 onward ride NEPRA's State-of-Industry Detail-of-Generation listings, parsed into the same panel (`scripts/ingest_sir_generation.py`). The SIR lane carries grid coal at evidence class TE2 (stated plant by plant) and gas at TE3 — the mixed-labelled RLNG/HSD combined-cycle group is assigned to gas by the annual control's own arithmetic, and an inferred fuel is never graded TE2. Coal and gas only, deliberately: the liquid fuels already ride OCAC's Total Power customer-class row with directly differenced months, and a generation shape must not displace a sales record — the SIR furnace-oil series serves as a published crosscheck against the OCAC profile instead. Years neither source covers are refused rather than fitted: FY2016–17 and FY2017–18 gas (PBS labels barely a third of gas generation in those years) and FY2021–22 gas (the annual-control bound misses the 95-percent rule by one point) stay flat with the refusal stated. Captive self-generation is deliberately not on this profile: a plant burning its own gas behind the meter follows its host industry, not national dispatch. The timing target is *generation*, and every month ships in the national monthly product with its class and basis on the row.

3.7. What this inventory has not obtained

Fuel quality is the sector's largest data gap. NEPRA's State of Industry Report publishes fuel quantities but no analysis—no sulphur content, no ash, no calorific value per consignment, and no split of the combined coal figure between imported and domestic. Nor is there any national record of which plants carry precipitators or flue-gas desulphurisation, or at what removal efficiency. Those absences shape Sections 4 and 8 respectively.

4. Methods

4.1. Grid generation: energy in, energy out

For each fuel f and gas g,

$$E_{g,f} = A_f \cdot c \cdot EF_{g,f}, \tag{1}$$

where A_f is the fuel consumed in tonnes of oil equivalent, c the TOE-to-terajoule constant, and $EF_{g,f}$ the IPCC Table 2.2 default. There is no efficiency term, no heat rate and no generation figure anywhere in the estimating chain; the plant ledger's generation data serve only the diagnostics of Section 6 and never enter Equation (1).

4.1.1. Why the unit choice is the method

The coal row hides an unpublished blend. Pakistan burns imported bituminous and domestic Thar lignite, and NEPRA reports one combined coal figure. IPCC's defaults for the two are:

Coal	kg CO_2/TJ	NCV (TJ/Gg)	kg CO_2/tonne
Other bituminous	94 600	25.8	≈ 2 440
Lignite	101 000	11.9	≈ 1 200
Ratio	**1.07**		**2.03**

Table 3: The same two coals, two units. Per terajoule they differ by 6.8 %; per tonne by a factor of two, because their calorific values differ by 54 % and lignite's own published range spans 5.50 TJ Gg^{-1} to 21.6 TJ Gg^{-1}.

An inventory reading tonnes of coal must assume a blend, and that assumption carries most of the sector's uncertainty. An inventory reading terajoules—which is what tonnes of oil equivalent already are—carries a 6.8 % residual instead.

The residual is not assumed away. The sector reports both bounds: all-bituminous gives 39.00 Mt and all-lignite 40.32 Mt, a span of 1.32 Mt around a central 40.04 Mt (the grid family's reference; alternatives 38.45 heat-rate, 44.05 with K-Electric, 57.82 legacy control) computed at a 79 % lignite share read from the plant fuel ledger (Section 5).

4.1.2. No separate oxidation factor

The 2006 Guidelines assume complete oxidation and fold it into the emission factor; Table 1.4 carries an explicit oxidation column of 1.0 for every fuel. Applying the 1996-vintage 0.98 on top would double-discount. This inventory's waste sector *does* carry an oxidation factor, because open burning genuinely is incomplete, and tests pin both conventions so neither is "corrected" into the other.

4.2. Captive gas self-generation: volume to energy to emissions

OGRA reports a volume and IPCC factors are per terajoule on a net calorific basis, so the conversion needs a heating value:

$$A = V \cdot 365 \cdot h \cdot \kappa \cdot \eta, \tag{2}$$

where V is MMCFD, h the gross heating value in Btu per standard cubic foot, κ the exact Btu-to-terajoule constant, and η the gross-to-net ratio.

Two of those are the assumption. Pakistani pipeline gas runs roughly 900 to 1000 Btu/scf depending on field and RLNG share, and nothing read here pins it; the utilities bill in MMBTU at a nominal 1000. Net calorific value is about 10 % below gross for natural gas, and omitting that step would overstate the category by 11 %, because IPCC factors are defined on the net basis.

At 1000 Btu/scf the category is 5.55 Mt; at 900 it is 4.99 Mt. Both are reported, as the cement sector reports both clinker ratios. Everything else in the chain is metered gas or an IPCC default.

4.3. Two categories, one module

Grid and captive generation are separate IPCC categories—1A1a and 1A2—computed in one module, as the transport sector carries road and off-road machinery together. They share the subject, the plant register and the FY2023–24 reference year; they do not share a factor table or a provincial proxy. Nothing downstream conflates them, because the national rollup, the key-category screen and the completeness table all work on categories rather than modules.

Both categories report FY2023–24: the grid from the 2025 edition's national fuel table, captive from the same fiscal year's metered gas—one reference year across the sector.

5. Emission Factors, and the Boundary of a Finding

5.1. Table 2.2 for the grid, Table 2.3 for captive

IPCC Volume 2 Chapter 2 publishes four parallel default tables for stationary combustion: energy industries, manufacturing industries and construction, commercial, and residential. Grid generation is 1A1a and takes Table 2.2. Captive self-generation serves the factory that owns it rather than selling power, so it is 1A2 and takes Table 2.3.

Natural gas	CO_2	CH_4	N_2O
		kg/TJ	
Table 2.2, energy industries	56 100	1	0.1
Table 2.3, manufacturing	56 100	1	0.1
Table 2.5, residential	56 100	5	0.1

Table 4: For natural gas, the industrial and energy-industry tables are identical. Only the residential CH_4 differs.

Captive gas takes Table 2.3, the manufacturing-industries table. For natural gas its factors are identical to Table 2.2's, so the choice carries no numerical weight here; it is cited correctly because the citation is what an auditor checks, and because it will matter if the captive fleet later resolves into oil- and coal-fired plants, where the tables do differ (Ahmad, 2026).

5.2. The coal blend as a parameter, not a decision

The lignite share is 79 %, read from the plant fuel ledger: the Thar-named mine-mouth plants are 75.2 % of FY2023–24 coal generation and the imported-coal plants 16.4 %, with one residual—Lucky Electric, at 8.4 %, whose name states no origin. The residual is carried as bounds rather than assigned: 0.752 with Lucky as imported, 0.835 with Lucky as Thar, and 0.79 is their midpoint. An independent route—plant coal mass weighted by IPCC rank calorific values—gives 0.70 to 0.84, overlapping the generation route, which is what makes the range defensible without resolving the plant. The heat-rate channel of Section 6.6 has since produced evidence on the plant itself: Lucky's implied calorific value sits at 11.4 to 12.1 TJ/Gg in all three of its years—lignite, not imported bituminous—which favours the upper bound. The bounds are kept until a source states the origin outright, but the residual is no longer evidence-free.

This replaces an inference from an older fleet. The previous value of 35 % was read off an earlier year's generation mix, before Thar Energy, ThalNova and Lucky entered service; by the anchor year the mine-mouth plants dominate. The parameter now carries 12 % uncertainty, spanning the union of both routes.

Because the calculation is on an energy basis, that uncertainty propagates to about 1 % on the coal factor rather than to a factor of two. This is the practical payoff of Section 4, and the blend remains a parameter with a published range rather than a decision the compiler must defend.

5.3. The TOE conversion, which is this sector's largest unresolved risk

The tonnes-of-oil-equivalent constant is 0.041 868 TJ per tonne, cited to the Eurostat definition—a *net* calorific basis, matching the IPCC factors. This is not an IPCC value; the 2006 Guidelines work in TJ/Gg and never define TOE, so it must be cited elsewhere.

The risk is that NEPRA's own TOE convention is unverified. If the regulator converts on a *gross* basis, as some South Asian statistical practice does, this sector overstates CO_2 by roughly 5 % for coal and oil and 10 % for gas. A one-sided conversion error cannot be expressed as a symmetric sensitivity without implying a symmetry that does not exist—so it is not folded into any ± percentage. It is instead a named structural case: the gross-basis scenario computes to 37.17 Mt against the 40.04 family reference (alternatives 38.45 heat-rate, 44.05 with K-Electric, 57.82 legacy control) (Table 8), with its own conditional interval and no probability attached. The evidence leans net—the MMBTU-metered plants imply 48.7 % efficiency, which only a net basis supports, and Pakistan's CRT filing states NCV outright—but evidence is not documentation.

Resolving it requires one page: the conversion-factors table at the front of the HDIP Pakistan Energy Yearbook. It is recorded as this sector's highest-priority acquisition.

6. Validation

6.1. Pakistan's own UNFCCC filing, cell-audited

Pakistan filed Common Reporting Tables with the UNFCCC in 2024, for inventory year 2022. They are the official statement of what the country reports for this category, and every figure below is re-derived from the workbook's cells on each run, with the filing's arithmetic (activity × implied factor = reported emissions) reproduced exactly before anything is compared.

The filing reports 1.A.1.a at 777 243 TJ and 62 841 kt CO_2 for calendar 2022, and three of its cells matter more than its total.

The basis column reads NCV. Every 1.A.1.a row states the net calorific basis—the official answer to what Section 6.5 approached physically. The sector's largest declared risk is now covered from two independent directions: the plant meters imply an efficiency only a net basis supports, and the country's own filing states the basis outright.

Gaseous fuels are not estimated. The official public-electricity category carries NE for gas, in a country whose regulator reports gas and RLNG as 58 % of thermal fuel energy—306 000 TJ in this inventory's matched year with no official counterpart. The filing's liquid-fuel activity is several times the regulator's fuel oil and diesel combined, so a reallocation may be at work, but as filed the category either omits nearly half the fleet's fuel or books it unlabelled. This is the kind of finding a cell-level official comparison exists to surface, and no model comparator could have.

The solid-fuel line corroborates two numbers here. The CRT's 324 724 TJ agrees with the regulator's coal energy to 2 %—two institutions, one answer, which cuts against the reading of the coal-NCV diagnostic in which the regulator's coal energy is overstated by most of a factor of two. Section 6.6 now sets a third channel against both: the per-plant heat rates side with the diagnostic, leaving this agreement as the counter-evidence rather than the settlement. And its implied CO_2 factor of 97.57 t/TJ sits at a 46 % lignite share on IPCC endpoints: between this paper's superseded 35 % and its ledger-derived 79 %, from 2022 when the Thar fleet was mid-ramp—consistent with a fleet in transition.

The years overlap by half (calendar 2022 against July 2022–June 2023), so magnitude comparisons bound rather than test; the fuel-scope and basis findings above do not depend on the year at all.

6.2. Grid generation against EDGAR

EDGAR reports Pakistani power industry emissions at 56.80 Mt CO_2e for 2024 (Crippa et al., 2025). This inventory reports the family reference 40.04 Mt (alternatives 38.45 heat-rate, 44.05 with K-Electric, 57.82 legacy control) for FY2023–24, or 70 % of it—different years, so the comparison bounds rather than tests. That ratio was close to unity on the legacy control and the direction of the change is not evidence either way: EDGAR's power figure for Pakistan is built from the same international energy balances that feed the aggregate fuel control this inventory stopped booking.

EDGAR remains the external benchmark; the official comparison above is the primary one. The taxonomy match is close: EDGAR's power industry category is essentially 1A1a, built from the same kind of national fuel statistics, so the comparison is nearly like-for-like rather than approximately so. Agreement at 142 % between two independently constructed estimates, in different reference years, is about as close as this class of comparison gets.

The residual is plausibly captive and industrial self-generation, which NEPRA's scope excludes and which EDGAR's power category may partly include. That reading is testable now in a way it was not before: this inventory's captive category is 5.55 Mt, so if EDGAR's figure included all captive gas generation the two would be further apart, not closer. The more likely reading is that EDGAR includes some self-generation and this inventory's grid figure and EDGAR's differ by less than either's uncertainty.

6.3. *Captive gas self-generation against EDGAR's industrial combustion*

EDGAR reports Pakistani industrial combustion at 43.98 Mt. This inventory reports 5.55 Mt against it, a ratio of 0.13.

That ratio measures *coverage*, not agreement, and the distinction matters enough that the benchmark row says so explicitly. This category estimates one fuel in one use: gas burned by industrial plants generating their own electricity. It does not estimate industrial process heat, boilers, kilns, captive plants burning oil or coal or bagasse, or the kiln fuel behind the cement category. A ratio near 0.13 is therefore the expected result rather than a discrepancy to investigate, and a ratio near 1.0 would indicate an error.

The internal reconciliation is the one that has teeth. The plant register sums to 288.01 MMCFD against OGRA's independently published 288—two sources built from different records agreeing to 0.01. That is what establishes the category is describing Pakistan's captive fleet rather than a subset or a superset of it.

6.4. *The time series as an internal check*

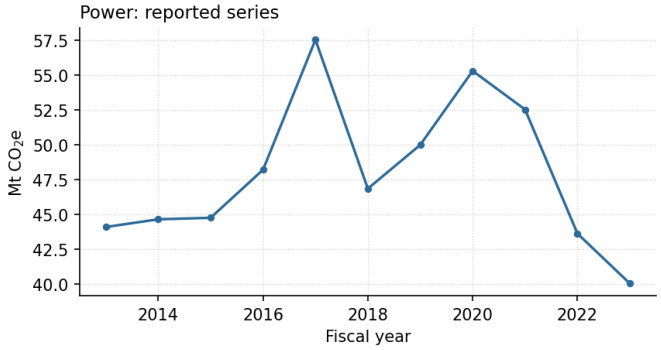


Figure 2: Grid emissions FY2013–24. The furnace-oil era's climb, its one-year collapse in FY2018–19, the coal-era rise to FY2021–22 and the partial recovery track the fleet's own history; the added early years parse from five earlier report editions, each reconciled against its own printed totals.

The series now runs eleven years, FY2013–14 to FY2023–24: 44.09, 44.64, 44.75, 48.19, 57.52, 46.84, 49.98, 55.28, 52.51, 43.62 and 40.04 Mt (the family reference; alternatives of three kinds: 38.45 heat-rate, a fuel-property alternative; 44.05 with K-Electric, a coverage alternative; 57.82 legacy control, the previously published account)—extended backward in this revision by parsing five earlier report editions, every added year reconciling against its edition's own printed Total row, with overlapping years reproduced identically across editions. The shape is a fleet history: a climb through the furnace-oil era to the FY2017–18 peak, a 10.7 Mt fall in one year as oil retired, and the coal-era climb to the FY2021–22 peak. Movements that large correspond to known fleet and demand events rather than to smooth interpolation, which is what a series built from a modelled activity proxy would produce.

What those movements were is now decomposed rather than narrated. An additive LMDI-I decomposition over the fuel table—exact by construction, its terms closing to machine precision—attributes the −4.1 Mt change from FY2013–14 to FY2023–24 as −5.2 Mt of activity (less fuel energy booked), 0.9 Mt of mix and 0.2 Mt of factor—and the mix term is close to a pure fuel swap: coal contributes 21.0 Mt and furnace oil −24.2, with gas at 4.8. A decade of Pakistani power-sector change is, in megatonnes, one fuel replacing another.

The direction of the total changed with the coal re-base, and the reason is the activity term. While the sector booked NEPRA Table 3's coal energy this decade read as a rise, and this paragraph described the activity term as *more* fuel energy burned. Booking the generating stations' own metered coal at default rank calorific values instead lowers the reference year's coal energy by 46 % and FY2013–14's by only 36 %, so the booked total energy falls across the decade where the printed conversion had it growing. The fuel swap is unaffected—it is a share, not a level—and the coal and furnace-oil mix terms are what they were. What moved is the level, which was the part Table 3's flat conversion was setting. The one year the story is not mix is FY2021–22, whose −2.8 Mt step was mostly activity (−3.2): a demand story rather than a fleet change, in either direction and on either coal account. The factor term is no longer zero. An earlier edition applied the FY2023–24 coal blend to every year, so the term vanished by construction, and this paragraph recorded that as a limitation and predicted the true series would carry a small positive factor term as the fleet moved from imported bituminous toward Thar lignite. The per-year blend is wired now—held in `data/power_coal_blend_by_year.csv`, zero before the Thar fleet existed and 0.79 at the reference year—and the prediction held at 0.2 Mt. The decomposition stays inside Table 3 deliberately; folding in the generation table would import the coal coverage mismatch of Section 6.5 and narrate an artefact as efficiency.

6.5. *The implied heat rate, and what it says about the activity tables*

The regulator's report contains two tables this inventory had never made talk to each other. Table 3 gives fuel consumed by fuel; Table 17 gives net generation by fuel and system. Dividing one by the other gives an implied heat rate—and a heat rate has physical bounds, so the ratio tests whether the two tables describe the same fleet.

Fuel	Implied efficiency	Spread	Verdict
Gas (incl. RLNG)	37.7 % to 56.0 %	×1.49	stable
Furnace oil	39.7 % to 52.1 %	×1.31	stable
Coal	34.6 % to 39.3 %	×1.14	unstable
Diesel oil	44.1 % to 50.2 %	×1.14	stable

Table 5: Implied net efficiency over the nine fiscal years both tables now cover, FY2015–24, on material years (above 1000 GWh).

Gas is the reassuring row, and the longer window makes it more so, not less. Its implied efficiency steps from 38 % to 40 % in FY2015–17 to 48 % to 56 % from FY2018 onward—exactly when the RLNG combined-cycle fleet entered service—and then holds. A ratio of two statistical tables that tracks a

known fleet transition and then sits stably at what combined cycles achieve on a net basis is behaving like real fuel energy over real generation, in both eras.

Coal is not, on the account this check refuted. Read against NEPRA Table 3's published coal energy its implied efficiency runs 58.4 % in FY2020—above any coal plant ever built, ultra-supercritical included—and 34.6 % in FY2023–24, a three-fold swing in four years. No fleet changes efficiency like that. The conclusion is that Table 3's coal row and Table 17's coal row do not cover the same set of plants, and that the mismatch moves between years.

Two things follow, and the second matters more than the first.

This does not affect the emissions estimate. The sector is driven by Table 3 alone. The coverage of Table 17 is irrelevant to Equation (1), and no number reported in this paper changes because of the finding. It is published because a reader comparing this inventory's implied generation against NEPRA's would otherwise discover the discrepancy unaided and have no way to tell which table this sector used.

It partly settles the sector's largest declared risk. Section 10 records that the TOE figures may be on a gross rather than net calorific basis, worth 5 % to 10 % on CO_2. Gas at a stable 49 % net is what a combined cycle does on a *net* basis; read as gross, the same fuel would imply about 55 %, which is at the very top of what any fleet average reaches. That is evidence for the net reading rather than proof of it. The risk stays declared—but it is no longer untested, which it was for as long as this check went unrun.

The comparison rests on one mapping worth stating. Table 3 has a single "Gas" row while Table 17 splits gas from regasified LNG, and the fuel table has no RLNG row at all—so the fuel behind Table 17's RLNG generation can only sit inside Table 3's gas figure, there being nowhere else in the table for it to be. A test asserts that no RLNG fuel row has appeared, because if a future edition adds one, the mapping would double-count instead.

6.6. Per-plant heat rates arbitrate the coal conversion

The heat-rate instability above asked for the plant-level tables, and the report carries them: Tables 20, 23 and 28 publish per-plant heat rates in Btu/kWh for five fiscal years, never previously parsed here. Table 28's footnote states the basis—LHV—so heat rate times net generation is net-calorific fuel energy, and that energy over the ledger's coal mass is a calorific value, plant by plant, with Table 3 nowhere in the loop.

Before trusting the channel, the gas plants test it, because for MMBTU-metered plants the ledger holds true fuel energy. Three behaviours emerge, all reported: three plants (Liberty Daharki, Sindh Nooriabad-I and -II) print net heat rates that reproduce the ledger's fuel-over-generation ratio to four digits—fully circular, the heat rate *is* the fuel table; the pipeline-gas plants Uch and Uch-II sit at 1.104 to 1.109 against the net-export heat rate, exactly the natural-gas HHV/LHV factor, so same meters converted to the LHV basis the footnote promises; and the RLNG combined cycles drift year to year, an independent channel. No coal plant is in the circular set: six of eight print year-varying operational heat rates, and Sahiwal and Thar Coal Block-I print flat design values (the latter with a net heat rate below gross, which no plant can do; the cells are kept as printed and flagged).

The result arbitrates the diagnostic. The Thar mine-mouth plants imply 10.7 to 13.9 TJ/Gg across their plant-years, mass-weighted 11.1 in FY2023–24—squarely the IPCC lignite value of 11.9—and the imported-coal plants 19.1 to 25.1, the bituminous band. The FY2023–24 fleet, mass-weighted at 99 % ledger coverage, implies 12.0 TJ/Gg against the 24.2 that Table 3's coal energy requires: a factor of 2.01, and 1.62 in FY2022–23 at full coverage. Whether the recovered figures are physical calorific values or the values NEPRA's own reporting chain assumes is undecidable from this document—and does not matter, because either reading pins what the official system believes Thar coal's calorific value to be, and both impeach a conversion that treats the fleet's coal at 24.2 TJ/Gg. Two plant-years fail a sanity screen and are excluded as outliers with their inputs named (Sahiwal FY2021–22, where the ledger's mass cell is suspect; China Power Hub FY2023–24, a 131.77 kt partial year). As a bonus, Lucky Electric—the ledger's one unresolved coal origin—implies 11.4 to 12.1 TJ/Gg in all three of its years: lignite-band evidence that the blend's true value sits near the upper bound of Section 5's range.

Two channels now disagree with Table 3's coal row and one agrees with it. The plant heat rates and the rank-default diagnostic both put the fleet's energy near half the official figure; the CRT's solid-fuel line agrees with the official figure to 2 %. "A scope difference only NEPRA can close" was this paper's whole statement of the problem for three releases; the candidate scopes are now enumerated and scored in `outputs/power_table3_hypotheses.csv`, one row per reading with the held evidence that bears on it. Of the seven: three are *consistent* with everything held (an imported-bituminous calorific convention applied to a lignite fleet—the implied 24.2 TJ/Gg sits inside the imported-only band whose mass-weighted value is 22.99—a gross-basis TOE column, priced at 37.168 Mt and able to explain only the 0.90 step, and Table 3 being right as published, the CRT's reading); one is *discriminated against* (K-Electric's ledger fuel is gas and furnace oil, so its boundary cannot carry a coal factor of two); and three are *untestable with held sources* (coal stock change, non-power coal inside the row's scope, and output-derived heat rates—the circularity screen covers 82 plant-fuel comparisons, none of them coal). The register selects nothing: the estimate keeps Table 3 as its control total under the family's declared selection rule; the restated case—ledger mass times per-plant implied calorific value, 190 737 against 206 606 TJ—is published as a labelled sensitivity of 38.45 Mt for the grid category, the largest labelled case in the totals table. The Energy Yearbook's conversion-factor note, a paid acquisition this project has not made, would decide between the leading consistent reading and the CRT's.

6.7. *Climate TRACE, plant by plant*

Climate TRACE reports 56.19 Mt CO_2e for the Pakistani assets it labels coal—2431 g kW^{-1} h on its own activity figures, against a physical ceiling near 1100 for coal combustion—where this inventory's coal is near 21 Mt. That gap has been regression-tested as a total; the per-asset file, joined to the point-source register, now resolves its structure.

Three findings. First, *every* coal-labelled asset carries exactly 2431 g kW^{-1} h, to the gram—including Fauji Fertilizer's and Pakistan Steel's captive units, which are not grid plants at all. One modelled intensity attaches to the label, not to any plant's operation. Second, three pairs of assets report identical activity and emissions to the megawatt-hour (Balloki/Bhikki/Haveli Bahadur Shah as triplets; ThalNova with Thar Energy; Lucky with Thar Block II)—an imputation signature, sitting exactly where the anomaly is largest. Third, the file carries misassignments (Hubco typed coal while the ledger records zero fuel burned in FY2023–24; two Dera Ghazi Khan Cement rows 570 km apart) and no year column, so every comparison bounds rather than tests. Against that: the exact-MMBTU gas plants agree with the engine at ratios of 0.97 to 1.03—the benchmark works where fuels are metered, and the anomaly is specific to the coal label's factor. That strengthens the standing reading—upstream Thar mine methane folded into plant rows, or a modelling artefact—over any reading in which this inventory is missing 36 Mt of coal combustion.

6.8. *Sentinel-5P sees the fleet*

Every comparator above ultimately rests on statistics someone reported. Sentinel-5P's TROPOMI instrument (Veefkind et al., 2012) does not: it measures NO_2 and SO_2 columns from orbit, daily, and NO_2's hours-scale lifetime keeps the signal near its source. From a daily 0.1° archive over Pakistan (2020–2026, quality-filtered GEE `S5P/OFFL/L3` composites of the Copernicus Level-2 product (Copernicus Sentinel-5P (processed by ESA), 2018); Copernicus Sentinel data, free and open with attribution), each major generating site's column enhancement over a same-day 0.3° to 0.7° annulus background was extracted for 2,275 days—nine sources, with complexes the grid cannot separate treated as one source each and labelled so—then aggregated to season-balanced fiscal years (300+ valid days behind most annual means). Four hypotheses were stated before the full run, each keyed to something the ledger already claims.

The Thar ramp is decisively visible. As the mine-mouth cluster went from one plant to four and 3.6 to 13.2 Mt of coal, its NO_2 enhancement rose 2.2-fold (FY2019–20 to FY2023–24) and its SO_2 enhancement—the lignite sulfur signature—9.0-fold, from 9.6×10^{-6} to 8.6×10^{-5} mol m^{-2}: the strongest signal in the study, and an independent witness to the build-out the LMDI decomposition quantifies from the fuel side.

The Hub site fades exactly as the ledger retires it. Hubco's furnace oil fell 977 kt to zero and China Power Hub's imported coal 2253 to 132 kt; the site's enhancement halves, FY2020–21 1.8×10^{-5} to FY2023–24 9.3×10^{-6} mol m^{-2} (ratio 0.53 against a pre-stated threshold of 0.65). The FY2024–25 rebound in the satellite series sits beyond the ledger's coverage and is reported as an observation, not a verdict.

The satellite arbitrates the Sahiwal FY2021–22 mass cell. The heat-rate channel flagged that year's ledger coal mass as suspect (1444 kt against 6881 GWh, implied CV 44 TJ/Gg). Generation says the year ran at 0.94 of the previous one; the suspect mass says 0.53. The satellite—which cannot read NEPRA's tables—says 1.02, siding with generation and independently corroborating a data-quality flag raised by an entirely different method. The following year, when generation genuinely collapsed, the enhancement fell to 1.3×10^{-6}: the instrument tracks the plant.

Rank agreement, and an honest control. Pooling within-source ranks across 36 source-years, fiscal-year enhancement orders the years the way ledger generation does: Spearman $\rho = 0.58$, $p = 2 \times 10^{-4}$ (the Thar cluster alone: $\rho = 1.0$). Uch, the steady field-gas null control, holds flat (CV 0.17). Guddu was drafted as a second null and the data rejected the premise—its fuel really does swing by a third across the window, and its enhancement follows—which is covariation evidence, honestly reclassified rather than a failed control.

The limits are declared with the channel. No wind rotation: plume smearing into the annulus biases every enhancement *low*, making the detections conservative. The Port Qasim complex sits inside Karachi's plume and supports only within-source time contrasts. The columns are daily L3 composites without averaging-kernel handling, and nothing here converts a column to a mass emission rate—this channel ranks and tracks; it does not weigh. Nothing in it feeds the estimate.

6.9. *The K-Electric leg, made decision-ready*

The boundary audit finds, in every year, that the plant sum brackets the national gas row only when the K-Electric system is excluded—which favours the reading that Table 3 omits K-Electric and is not proof of it. The furnace-oil rows point the other way in four of the five years, the sum with K-Electric sitting closer to the national row than the sum without it. The FY2023–24 furnace-oil year is the exception and the sharpest discriminator in the panel: the without-K-E plant sum sits at 0.890 times the national RFO row and the with-K-E sum at 1.256 times it, because K-Electric's furnace oil collapsed in the same year the national row did. Whether the national fuel statistic already contains K-Electric's fuel is therefore not established.

The whole ladder now ships as one table (`outputs/power_toe_reconciliation.csv`, rc30): per year and fuel, the national Table-3 energy (recovered from the audit's own ratio pair, so the table carries no calorific constant of its own), the CPPA plant sum, the K-Electric plant sum and the heat-rate-implied coal energy, side by side. It settles nothing the configurations leave open—settling needs HDIP's conversion-and-scope metadata, the one paid or written-answer item in this sector—but it puts every rung of the comparison in one place. When that metadata arrives, the family's names map onto the resolution the eighteenth review proposed: `inventory_reference` is the *hdip-toe-reported* reading,

`power_plant_physical_coal` the *plant-physical-NCV* reading, and `power_with_kelectric` the *hdip-toe-plus-KE* reading.

What that boundary question is worth is now computed rather than gestured at, and it is not the flat five-to-six megatonnes earlier versions of this paper carried. The K-E ledger rows imply 7.1, 7.7, 6.3, 4.9 and 4.0 Mt CO_2 across FY2019–24: a 48 % decline in five years, driven almost entirely by furnace-oil retirement (977 down to 209 kt) while gas collapsed into RLNG. Whichever way the boundary decision goes, the correction is a trend, not an adder; the coverage alternative is priced for all nine panel years in `outputs/pakemis_structural_trajectories.csv` (axis *power scope*). Two corroborations ride along: implied net efficiency rises from 36.9 to 45.5 % across the window—what RFO steam giving way to RLNG combined cycle should do, and a check that the volume-based heating-value assumption is not badly wrong—and the arithmetic reproduces the boundary audit's energies exactly.

The case is published as the coverage alternative "grid + K-Electric ledger fuel": 44.05 Mt against the 40.04 headline, priced beside the reference and not added to it, because adding it would count K-Electric's fuel twice if the national statistic already contains it. Its evidence classes are weaker than the CPPA leg's—volumes at an assumed heating value, oil at default calorific values, only the two Nooriabad units metered exactly—and that weakness, with the open boundary question, is what keeps it an alternative rather than a category.

6.10. What is not validated

Fuel quality. Nothing here tests the coal blend, the calorific values, the TOE convention or the gas heating value against a Pakistani laboratory measurement, because none is published in anything read here. Those four questions carry this sector's real uncertainty, and the uncertainty statement now says so in kind rather than in width: the blend and the heating value are parametric with ranges read off their corroborations, while the calorific and conversion questions are one-sided structural cases in Table 8, each with a conditional interval and no probability—because a ± percentage that absorbed a factor-of-two question would misdescribe both the question and the percentage.

Which basis the historical years should use.. NEPRA's per-plant generation panel covers FY2015–16 to FY2019–20 and the plant fuel ledger begins in FY2019–20, so exactly one fiscal year carries both, and that year decides which provincial vector the earlier years should take. Emissions follow fuel rather than generation, so the fuel vector is the control and the two candidates are the panel's own-year generation and the reference-year vector the release would otherwise borrow for every period. The comparison ships per stream and province as `outputs/pakemis_power_grid_basis_change.csv`. Measured as total-variation distance from that control, furnace oil's own-year generation lands at 0.005 against 0.066 for the borrowed vector; for gas the ordering reverses, and the borrowed vector is much the closer of the two. Generation shares equal fuel shares only where plants burn at a common heat rate: the furnace-oil fleet is near enough uniform for that to hold, and the gas fleet—Punjab's RLNG combined cycle against the low-BTU plants in Balochistan—is not. On that comparison furnace oil was allocated on own-year generation for the five years the panel reaches and gas was left on the reference vector, through release 1.0.0rc65. Neither is applied now. The comparison had been run on a panel holding under half of that year's gas generation, and the question it posed was the wrong one: the same report tables print each station's fuel quantity beside its generation, so the earlier years take the fuel itself. From rc66 every fuel is split between provinces by each year's own station fuel record (`outputs/pakemis_power_provincial_basis_by_year.csv`), coal by station tonnage at the calorific value the national account books it at; the comparison above remains as the record of how far generation sits from fuel.

7. Results

7.1. The sector

Category	IPCC	Mt CO_2e	±%
Electricity and heat production	1A1a	40.040	10.9
Captive gas self-generation	1A2	5.549	15.2
Total		**45.590**	**9.7**

Table 6: Electricity generation, both categories on FY2023–24. The uncertainty column is Approach 1 per category and for the sector, with the four grid fuel rows' activity errors treated as fully correlated through the shared TOE conversion (independently they would give 7.3 %); the one-sided questions—the coal conversion, the TOE basis, K-Electric coverage—are structural cases in Table 8, not width in this column.

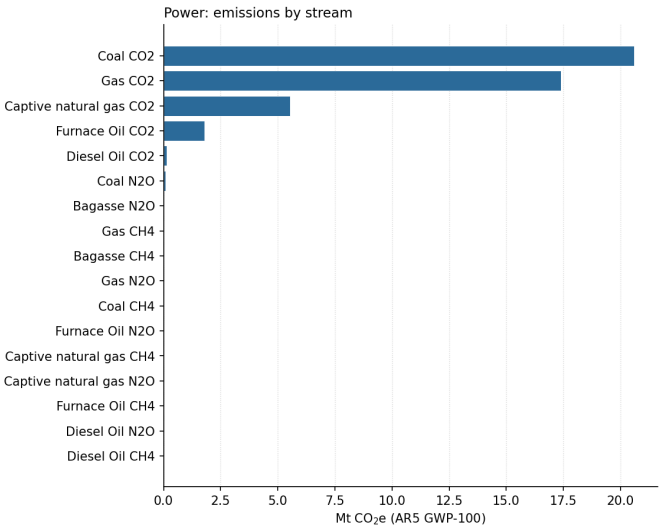


Figure 3: Emissions by fuel and gas. The sector is 99.6 % CO_2: the utility CH_4 and N_2O factors in Table 2.2 are small precisely because large boilers burn completely.

Coal is 20.59 Mt of CO_2, gas 17.40 Mt, furnace oil 1.79 Mt and diesel 0.13 Mt. Coal is 37.9 % of the energy and 51 % of

the grid emissions, which is the carbon-intensity difference between coal and gas expressing itself directly—and the reason a fleet shifting between them moves emissions faster than it moves generation.

The grid category now carries the sector's smallest stream and its first memo item. Eight sugar-mill cogeneration plants sold 726.85 GWh of bagasse-fired electricity to the grid in FY2023–24 (Table 8's E3 block, reconciled against its printed total); at IPCC biomass factors that is 0.02 Mt of CH_4 and N_2O—the one non-fossil methane in the sector, at AR5 28—and 1.09 Mt of biogenic CO_2 reported as a memo outside every total, the carbon having been fixed by the same season's cane. The activity is generation times an assumed 15 000 kJ kW^{-1} h heat rate, the sector's one generation-derived stream, and it is labelled structurally weaker than the metered-fuel streams for exactly that reason. Three plants list biomass and one lists furnace oil as its alternate fuel; no source read here publishes the co-firing share, so the furnace-oil caveat is named rather than split.

7.2. *Sensitivities*

Case	Mt CO_2e	What it bounds
All coal bituminous	38.995	lower bound, grid
Reported (79% lignite)	40.040	
All coal lignite	40.318	upper bound, grid
Coal at implied CVs	38.452	the conversion question, grid (Sec. 6.6)
Table 3 published energy	57.820	legacy published control, grid
Captive at 900 Btu/scf	4.995	lower bound, 1A2
Reported (1000 Btu/scf)	5.549	

Table 7: Both published assumptions, bounded. The coal blend spans 1.32 Mt on the family reference 40.04 (alternatives 38.45 heat-rate, 44.05 with K-Electric, 57.82 legacy control); the gas heating value spans 0.55 Mt on 5.55. The last row is the aggregate fuel control booked before the coal re-base, kept here so the blend span and the account change are not read as one quantity.

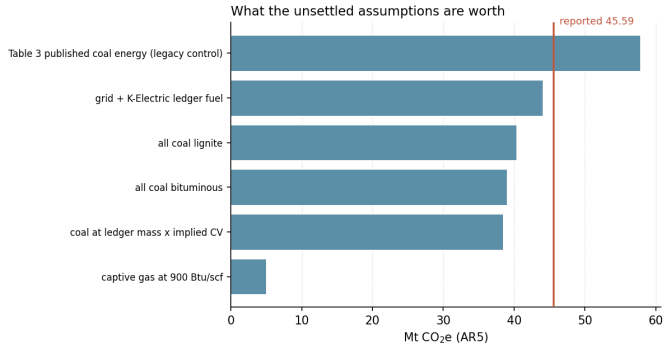


Figure 4: The same as a range around the reported figure.

Neither span is large relative to the sector, and that is the result. On a mass basis the coal row alone would span roughly a factor of two; on an energy basis it spans 3.3 % of the grid category. The account change is a separate and larger matter and is kept separate: the legacy control sits 17.78 Mt above the reference, which is 13 times the blend span, and no reading of the blend reaches it.

Structural case	Mt CO_2e	P2.5	P97.5
Reference (family reference)	40.040	36.0	44.5
All coal bituminous	38.995	34.9	43.4
All coal lignite	40.318	36.1	44.9
Coal at implied CVs	38.452	34.4	42.9
Table 3 published energy	57.820	51.8	64.3
TOE on a gross basis	37.168	33.3	41.4
Grid + K-Electric ledger fuel	44.054	39.8	48.5
Captive at 900 Btu/scf	4.995	4.3	5.8

Table 8: The one-sided questions as named cases, each with a conditional parametric interval and no structural probability. Discrete choices are kept unsampled so a confidence interval cannot conceal a boundary or coverage choice; the parametric ±10.9 % above is conditional on the reference configuration of every row in this table.

7.3. *Provincial allocation, corrected*

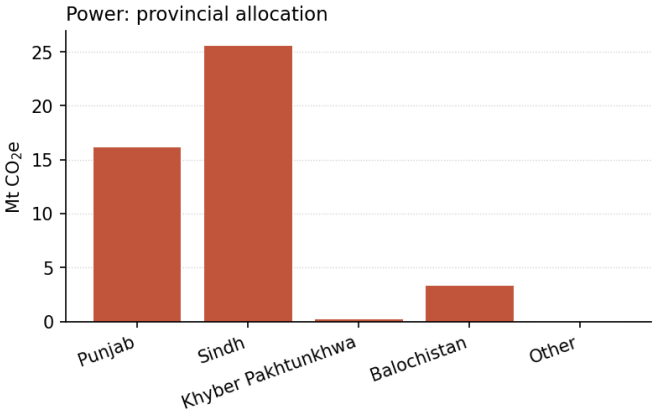


Figure 5: Punjab 16.24 Mt, Sindh 25.67 Mt, Balochistan 3.41 Mt, Khyber Pakhtunkhwa 0.27 Mt. Unlike almost every other sector in this inventory, the ordering does not follow population, because it follows the plants—and above all the Thar coalfield.

This geography has now been corrected twice, each time by a better class of evidence, and the second correction is larger than the first. The original capacity proxy gave Khyber Pakhtunkhwa 14.24 % of a combustion sector whose electricity there is hydro; plant-wise generation moved 6.2 Mt out of that province. But the generation proxy still blended all fuels onto one share, and the ledger joined to the report's own plant-location table (Table 5) shows the fuels do not share a geography: coal is 89 % Sindh on the stations' booked energy (four Thar mine-mouth plants, Port Qasim and Lucky against Sahiwal alone in Punjab), the RLNG combined-cycle corridor is Punjab, the pipeline-gas plants sit on the Sindh and Balochistan fields, and the furnace oil actually burned in FY2023–24 is entirely Punjab—Hub, the Balochistan plant that dominated older vintages of this split, reports zero fuel for the year. Allocating each fuel on its own plants moves 12.6 Mt from Punjab to Sindh at an unchanged national total and makes Sindh, not Punjab, the sector's dominant province.

Each per-fuel share is recomputed from the ledger by a test, so the committed geography cannot quietly outlive its derivation; the energy weights carry their evidence classes (MMBTU exact, volumes at the declared heating value, coal mass at the per-plant implied calorific values of Section 6.6). Appendix C records the replaced proxies.

7.4. Contribution to the national inventory

Category	Level (%)	Variance (%)
Electricity and heat production	9.00	0.80
Captive gas self-generation (1A2)	1.25	0.03

Table 9: Shares of the national total and of national Approach-1 variance.

Grid generation is the third largest category in Pakistan and—since the uncertainty rebuild—no longer a large contributor to national variance: 0.80 %, down from a published 3.24, and from an earlier vintage in which it ranked second after enteric fermentation. The fall is not new information about the fleet; it is the flat activity percentage being replaced by per-fuel widths the evidence supports, and the questions actually worth megatonnes—the coal conversion, the TOE basis, K-Electric coverage—moving to the structural-scenario table where a probability interval cannot conceal them. The honest summary is that this sector's parametric uncertainty is modest and its structural uncertainty is large, one-sided and named.

Captive gas self-generation contributes 0.03 % of national variance, which is the expected result for a category whose activity is metered and whose factor is an IPCC default: its 15.2 % band is small and its size is modest. Its value to the inventory is not uncertainty reduction but coverage—it is the first number inside a gap that was previously empty.

8. Air Pollutants: An Upper Bound, and a Re-Diagnosed Blocker

8.1. The blocker was fuel analysis, not emission factors

This sector's air-pollutant layer was previously recorded in this inventory's completeness table as not estimated, blocked on EMEP/EEA large-combustion factors that the agency no longer serves in a retrievable form. Working back to the primary tables showed that diagnosis was wrong, and correcting it is a more useful result than the layer itself.

US EPA's AP-42 gives no coal SO_2 factor as a fixed value. For bituminous coal, Table 1.1-3 publishes $38S$, where S is the sulfur weight percent of the coal actually burned; for lignite—which is what 79 % of this fleet's coal energy is—the separately fetched Section 1.7 publishes $30S$ (its own footnote scales the factor $22S$ to $34S$ with ash sodium, which is not held here, so the table's default is carried). Table 1.3-1 publishes $157S$ for No. 6 fuel oil. Filterable particulate is $10A$ (bituminous, Table 1.1-4) and $6.5A$ (lignite, Table 1.7-4), functions of ash content.

The factors were never missing. What is missing is the *fuel analysis* they scale with, and NEPRA publishes none. That is a different problem with a different solution: no amount of searching for factor tables would have closed it, and the earlier note would have sent a reader looking in the wrong place.

8.2. Fuel quality as a published range

Published Thar lignite sulfur spans 0.26 % to 2.90 %—a factor of eleven. Ash is better constrained, with two published means at 6.42 and 6.83 %.

Sulfur and ash are therefore carried as ranges, with the central value the geometric mean and the uncertainty read off the range—the same treatment this inventory's fugitives sector gives IPCC's published factor ranges, using the same function so the two cannot drift apart on the convention. The Thar ranges belong to the lignite share by construction; the imported bituminous burned at Sahiwal, Port Qasim and Hub has no consignment analysis in anything read here, so the bituminous share carries the same ranges as a *declared proxy*—imported bituminous is typically lower-sulfur than this range's centre, so if anything the bituminous side of the bound is generous, and the proxy is recorded in the fuel-quality file rather than silently reused.

8.3. The coal rank split, and why it raised the bound by half

Coal was previously one stream on the bituminous section. Since the Thar-lignite split, 79 % of the coal energy (`coal_lignite_share`, generation-weighted from the plant fuel ledger, ±12 %) rides Section 1.7's own factors and lignite's IPCC net calorific value of 11.9 TJ Gg^{-1}, the remainder Section 1.1's on 25.8. The rank split moves the bound substantially *upward*: a tonne of lignite carries less than half the energy of a tonne of bituminous coal, so a per-tonne factor scaled by the same sulfur or ash nearly doubles per terajoule—the same arithmetic through which the greenhouse-gas layer's rank blend works, now applied where it bites hardest. The firing configuration is a declared assumption (pulverized dry-bottom tangential, the configuration whose NO_x factor Section 1.7 rates A); no Pakistani document read here states the Thar boilers' firing arrangement. Lignite CO is ND in Table 1.7-1 for this configuration and is therefore absent rather than zero-filled. Grid bagasse cogeneration enters the layer from Section 1.8 on the same fuel energy the greenhouse-gas layer burns—uncontrolled filterable PM and NO_x only, because the section publishes no SO_2 or CO factor for bagasse, whose sulfur it describes as characteristically negligible.

8.4. The result, labelled as what it is

These are uncontrolled factors, and that word carries the whole caveat. AP-42's uncontrolled rows describe what leaves the furnace, not what leaves the stack. Pakistan's supercritical coal units were built with electrostatic precipitators and the newer ones with flue-gas desulphurisation, which remove most of the particulate and much of the sulphur. No national dataset records which control is fitted where or how it performs.

Pollutant	kt/yr	AP-42 source
PM (filterable)	362.3	1.7-4 (6.5*A*), 1.1-4 (10*A*), 1.8-1, 1.4-2
SO_2	208.7	1.7-1 (30*S*), 1.1-3 (38*S*), 1.3-1 (157*S*), 1.4-2
NO_x	113.4	1.7-1, 1.1-3, 1.3-1, 1.8-1, 1.4-1/3.1-1
PM_{10}	118.3	1.7-7 (2.3*A*), 1.1-4 (2.3*A*), 1.4-2
$PM_{2.5}$	34.3	1.7-7 (0.66*A*), 1.1-6 (0.6*A*), 1.4-2
CO	11.9	1.1-3, 1.3-1, 1.4-1
VOC	0.7	1.4-2

Table 10: Grid only, FY2023–24 fuel, **uncontrolled**—an uncontrolled screening scenario, each coal rank on its own AP-42 section since the Thar-lignite split. Captive self-generation is IPCC 1.A.2 and is priced once, in the industrial air layer (engine and turbine factors), so it no longer appears here. Applied to the same terajoules the greenhouse-gas layer burns, so both inventories describe the same combustion.

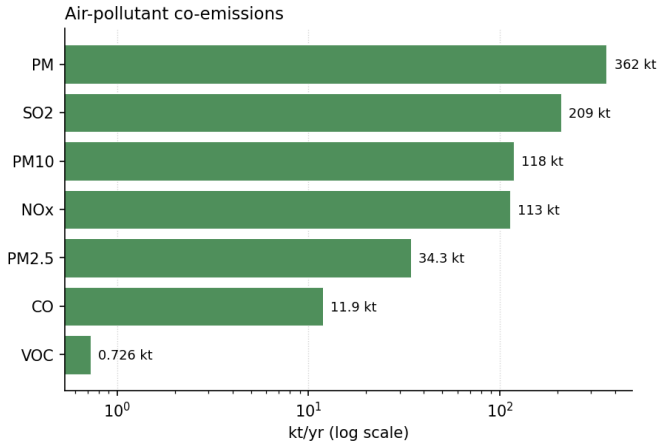


Figure 6: The same figures on a log axis, which the range demands.

Applying a removal efficiency would therefore mean inventing the number that matters most—it spans an order of magnitude between uncontrolled and well-controlled. The bound is reported as a bound, and the estimate of what the fleet actually emits is left not estimated, with the missing parameter named. A number labelled as a bound is useful; the same number labelled as an estimate would not be.

Where that parameter lives is now established rather than guessed. All five of NEPRA's Licensees HSE Performance Evaluation Reports (2021 through 2024–25) were obtained and audited for this revision: they publish no per-plant control equipment, emissions or measurements—they are occupational-safety scorecards, one aggregate points total per company—but their evaluation questionnaire scores licensees on "pollution control measures related to air quality" and on whether an environmental monitoring report was submitted. The controlled-emissions data this layer needs therefore exists at the regulator as unpublished licensee submissions under the Power Safety Code, and the concrete route to it is a request to NEPRA's HSE department, not a further search of the published record.

8.5. $PM_{2.5}$ from the size distribution, with its own check

An earlier version of this layer carried no $PM_{2.5}$ row, deliberately: AP-42 Table 1.1-4 gives filterable PM and PM_{10} for coal but no $PM_{2.5}$, and reporting the gas figure alone—about 1 kt, because AP-42 notes all gas particulate is under a micron—would have read as power being negligible for fine particulate against residential biomass at 282 kt under that sector's reference configuration, which uncontrolled coal is not.

The missing piece was the cumulative particle-size distributions—and each rank has its own. Table 1.1-6 (bituminous, dry-bottom pulverised) puts 6 % of filterable mass at or below 2.5 µm: a cumulative factor of 0.6*A* lb/ton, which the table rates C—below the A of the mass factor it scales, as a derived number should be. The same distribution puts 23 % at or below 10 µm, 2.3*A*, exactly the PM_{10} factor Table 1.1-4 already supplied; the two tables reproducing each other is the check on the transcription. Lignite's own Table 1.7-7 is coarser-shifted but sits on a heavier per-terajoule mass: 10 % at 2.5 µm (0.66*A*) and 35 % at 10 µm (2.3*A*) of the 6.5*A* tangential filterable factor. The result is 34.32 kt of uncontrolled $PM_{2.5}$, 33.31 of it from coal—roughly double the pre-split figure, because lignite's tonnage per terajoule doubles the ash-scaled mass—carried with the same bound-not-estimate caveat as every row above—and for fine particulate that caveat bites hardest, because electrostatic precipitators remove fine mass least efficiently exactly where the health burden concentrates.

8.6. A convergence that dissolved, which is the more instructive record

The consolidated plant workbook that supplied this paper's captive register computes its own grid SO_2 for 2024, on a generation basis with kilogram-per-megawatt-hour factors, at 253.2 kt. Before the rank split, the AP-42 fuel-basis figure sat within within a few per cent of it, and an earlier edition of this section recorded that agreement with the warning that it was a coincidence of assumptions rather than corroboration—both routes embedded the same all-bituminous reading of the coal fleet. The rank split proved the warning right by dissolving the agreement: on each rank's own section and calorific value the figure is 208.7 kt, 17.6 % below the workbook—it read 399.4 kt until rc60, when the oil and diesel share of this screening narrowed to the fuel the plant-level admission register does not price, and higher again until the coal re-base, which moved it to the other side of the workbook. That reversal is the same point made twice: this screening is coal energy times a sulphur content, so it tracks whichever coal account is booked, while the workbook's 253.2 kt does not move at all. The agreement was never evidence and neither is the disagreement's direction. The divergence now *measures* the shared assumption the old agreement concealed. Two estimates that guessed the same missing parameter agreed; correcting the parameter on one side moved it, which is exactly what agreement-by-shared-guess predicts and corroboration does not.

8.7. Gas NO_x: the ledger settles the combustor question, with a residual

AP-42 Chapter 1.4 is an *external combustion* chapter—boilers—while much of Pakistan's gas fleet is combined-cycle, which is Chapter 3.1, and the two uncontrolled NO_x factors

differ: 2.7×10^{-1} lb/MMBtu for a large wall-fired boiler against 3.2×10^{-1} for a gas turbine, both rated A. An earlier version of this layer called gas NO_x its weakest row because NEPRA's national fuel table reports fuel without plant type. The plant ledger changes that: it carries heat rates, and a plant at or below 9000 kJ kW^{-1} h net—40 % efficiency—can physically only be a combined cycle. Heat rate is trusted over names, because names lie here: Bin Qasim TPS-II and TPS-III are combined-cycle blocks behind steam-era names, and Guddu's units 14–16 carry a heat rate no combined cycle designs to. Names decide only where they assert turbine hardware (GTPS, CCPP), which heat rate cannot separate from a steam plant at simple-cycle efficiencies.

On FY2023–24 energy the grid gas fleet classifies as 80.2 % turbine, 13.7 % boiler, and 6.1 % unresolved—three plants in the 9,000–10,500 band, where a degraded combined cycle and a steam plant are indistinguishable from the ledger. The grid gas NO_x factor is the energy-weighted blend, with the unresolved share kept on the boiler factor and reported here rather than assigned. Because the turbine factor is the higher one, resolving the fleet *raises* gas NO_x by about 13 % over the all-boiler assumption; the total above moves accordingly. Only NO_x is split: the CO factors of the two chapters are numerically indistinguishable, and sulphur passes through to SO_2 regardless of how the gas is burned. Captive self-generation has left this table: IPCC 2006 books autoproducer generation under 1.A.2, and the industrial air layer prices it on stationary-engine and gas-turbine factors under declared fleet shares, reported once under industry.

9. Discussion

9.1. Choosing the unit is choosing the uncertainty

The coal blend is unknown, and how much that matters is entirely a function of the unit the activity data are read in. Per tonne it is a factor of two; per terajoule it is 6.8 %. Nothing about Pakistan's coal fleet changed between those two statements.

The general form is worth stating because it recurs. When an activity datum is available in more than one unit, the right one is the unit in which the unknown property does the least work—and calorific value is exactly the property that distinguishes the two coals, so reading energy rather than mass removes it from the calculation rather than requiring it to be assumed.

This has a corollary that cuts against the paper's own method. The TOE constant is itself a calorific-value convention, and if NEPRA reports on a gross basis rather than a net one, the sector is overstated by 5 % to 10 %. So the energy basis has not removed the calorific-value problem; it has moved it from a per-fuel blend question to a single documentation question about one constant. That is a better place for it—one page of one yearbook resolves it—but it is not an escape, and Section 10 treats it as the sector's largest unresolved risk rather than a footnote.

9.2. Coverage against accuracy

Captive gas self-generation contributes 1.25 % of national emissions and 0.03 % of national inventory variance. By the usual criterion for where inventory effort should go—reduce the variance—it was not worth building.

It was worth building for a different reason, and the distinction is worth naming. This inventory reports 89 % of EDGAR's Pakistani sectors summed, and the shortfall is not spread thinly across sectors but concentrated in categories that are entirely absent. Absent categories contribute nothing to variance precisely because they contribute nothing at all, so a variance-based priority ranking is blind to them by construction.

Coverage and accuracy are different objectives and need different criteria. The key-category analysis tells you where to improve what you have; the completeness table tells you what you do not have. Captive gas self-generation moved 5.55 Mt from the second column to the first, and no variance ranking would ever have recommended it.

9.3. On labelling a bound as a bound

The air-pollutant layer could have been published as an estimate. Applying a plausible removal efficiency—say 99 % for particulate and 90 % for sulphur where desulphurisation is installed—would have produced numbers that look like every other published power-sector air inventory and would have been impossible for a reader to distinguish from a measured result.

We publish the uncontrolled figure and label it, because the removal efficiency would have been the largest single term in the calculation and entirely invented. The resulting number is less immediately useful and more honest: it bounds what the fleet can emit, and it names the control-device inventory as the specific missing input.

The same principle once produced a decision *not* to report $PM_{2.5}$: with only a gas factor in hand, a gas-only figure would have been a real number, correctly computed, and would have misled every reader who compared it against residential biomass. An absent number is recoverable; a misleading present one is not.

That decision has since been reversed, and the reversal is worth recording rather than quietly deleting. The blocker was never that AP-42 lacks a coal factor—it was that we had not read far enough into the chapter. Table 1.1-6 gives the particle-size distribution for bituminous and subbituminous combustion, Table 1.7-7 lignite's own, and applying each to its rank's filterable mass yields 34.32 kt of $PM_{2.5}$ (Section 8). This is the same not-absent-but-unread failure the transport and residential sectors each hit, and the same correction: the honest response to a missing factor is to keep looking for it, and only then to declare it missing.

9.4. Implications

Read one page of the HDIP yearbook. The TOE convention is worth 5 % to 10 % of this sector—more than the coal blend, more than the gas heating value, more than any methodological refinement available—and resolving it requires

a conversion-factors table that exists and is paywalled rather than a measurement campaign.

A control-device inventory would convert a bound into an estimate. Which plants carry precipitators and desulphurisation, and at what efficiency, is knowable from environmental approvals and operator records. It is the difference between an air-pollutant layer that can be cited and one that can only be bounded.

Captive gas self-generation is a template for the rest of 1A2. The method here—find the regulator who regulates the *fuel* rather than the plant, and reconcile against an assembled register—closed 13 % of the industrial-combustion gap with metered data, and this inventory has since applied it to industrial process gas and to the industrial liquid fuels metered by OCAC's Total Industry customer class, taking the covered share to about a third. The remaining 16 Mt is specifically solid: industrial coal, petroleum coke and bagasse, in boilers, kilns and the captive plants that burn them rather than gas. The same approach applied to those fuels' suppliers would close a substantial further part without waiting for the paywalled yearbook.

Provincial allocation deserves the same scrutiny as the total. The capacity proxy this paper replaced was wrong by 7.9 Mt in one province and had been flagged as probably wrong, in the right direction, in the parameter file. That it survived until plant-wise generation became available says something about how rarely allocation proxies are revisited relative to how often totals are.

10. Limitations

The TOE convention is unverified, and it is the sector's largest unresolved risk. The conversion is applied on a net calorific basis to match the IPCC factors, but NEPRA does not state its own convention. If the regulator reports on a gross basis, this sector overstates CO_2 by roughly 5 % for coal and oil and 10 % for gas—larger than the coal blend, and one-sided, so it cannot be expressed as a symmetric sensitivity. Resolving it needs the conversion-factors page of the HDIP Pakistan Energy Yearbook.

The coal blend is plant-resolved but not fuel-analysed. The 79 % lignite share is read from the plant ledger's origin split, with Lucky Electric carried as bounds (0.752 to 0.835) rather than assigned—though the heat-rate channel now puts Lucky's implied calorific value in the lignite band. What no Pakistani source in this inventory provides is a laboratory fuel analysis: Table 3's own coal energy implies 24.2 TJ/Gg over a fleet that is 92 % lignite by mass—a ratio of 1.86 against rank defaults, and a factor of 2.01 against the per-plant implied calorific values of Section 6.6. Two channels against one (the CRT's solid-fuel agreement), published as a control-total finding with a labelled sensitivity rather than resolved. The energy basis keeps the blend's consequence to 3.3 % of the grid category; the conversion question itself is worth 17.78 Mt on the grid category—the distance between the booked station account and the aggregate conversion retained beside it—and is this sector's largest number at stake.

Captive gas self-generation is one fuel in one use. It covers gas burned for industrial self-generation. Captive plants burning furnace oil, diesel, coal or bagasse are in the register but have no metered fuel series; the liquid fuels are now estimated from OCAC's Total Industry customer class, but solid-fuel process heat, boilers and kilns are not estimated at all. With industrial process gas and the metered industrial liquids now estimated separately, roughly 16 Mt of EDGAR's 43.98 Mt for 1A2 remains outstanding, and this category is a floor rather than an estimate.

The captive gas heating value is unpinned. Nothing read here states it; the 900 to 1000 Btu/scf range moves the category by 0.55 Mt. Both are reported.

Air pollutants are an uncontrolled bound, not an estimate. No national control-device inventory exists, so the removal efficiency—the largest single term for particulate and sulphur—is not applied rather than guessed. Every pollutant here is therefore what the fleet would emit with no abatement, which is an upper bound and is labelled as one; what the fleet *actually* emits remains not estimated, and the completeness table says so.

Fuel sulphur and ash are literature ranges, not Pakistani measurements. Thar sulphur spans a factor of eleven in the published analyses, and the imported bituminous fraction is a different coal that NEPRA's combined figure does not separate.

Gas NO_x uses boiler factors for a partly combined-cycle fleet. AP-42 Chapter 1.4 covers external combustion; turbine factors are in Chapter 3.1 and differ. NEPRA reports fuel without plant type.

The anchor year rests on one edition's OCR-repaired column. The FY2023–24 national fuel figures come from the 2025 report, whose text layer renders some thousands separators as decimal points; the repair is validated against the overlapping years of the 2024 edition and the column's own printed total, but it is a repair.

The grid series starts at FY2013–14. Five earlier report editions were parsed to extend it backward in this revision, each reconciling against its own printed totals, but editions before FY2013–14 do not carry the fuel table in extractable form, so the sector still cannot contribute to a national trend assessment reaching a 1990 base.

Captive self-generation before FY2016 is not covered, and two years within the series carry quality flags: FY2021 is provisional, read by OCR from a chart label in an image-scan edition, and FY2023's SSGCL figure was restated upward in the following edition. Neither is the reported year, and both are carried with their flags rather than smoothed.

11. Conclusion

Pakistan's electricity generation emitted 45.59 Mt CO_2e in FY2023–24: 40.04 Mt from grid generation (the selected member of the structural family whose alternatives read 38.45, 44.05 and 57.82 Mt) and 5.55 Mt from industrial captive gas self-generation, together 10.2 % of a ten-sector national inventory, on a single reference year. Beneath those two numbers sits a

plant fuel ledger covering 99.0 % of thermal generation, validated per fuel against a table its parser never reads—and the paper's results are as much about what that ledger reveals as what it sums to.

Four results generalise.

A ledger under a control total answers questions neither can alone. The national fuel table sets each fuel's magnitude; the plants beneath it set the structure. The coal blend stopped being an inference (35 % lignite, ±40 %) and became an observation (0.79, with the one plant whose name states no origin carried as bounds, 0.752 to 0.835). On the energy basis this activity is kept in, that residual moves the coal factor by about 1 %; per tonne it would be worth a factor of two.

Reconciliation is a test of the statistics, not only of the inventory. Forcing the plants and the national table to agree produced two findings about Pakistan's official energy accounts. The plants metered in MMBTU—already energy, no calorific assumption—imply 48.7 % net efficiency, evidence that the tonnes-of-oil-equivalent table is on the net calorific basis IPCC factors require, which had been this sector's largest declared risk. But the same table's coal energy implies 24.2 TJ/Gg over a fleet 92 % lignite by mass—and the report's own per-plant heat rates, parsed here for the first time, put the Thar plants at 11.1 TJ/Gg, squarely IPCC lignite, a factor of 2.01 against the national conversion; the plant sums bracket the national gas row only when the K-Electric system is excluded. Each is worth megatonnes, each is published with the evidence rather than absorbed into an adjustment. The first of the two is now DECIDED rather than carried: the sector books *grid_stations_rank_default* 40.04 Mt, the generating stations' own metered coal at default rank calorific values, because divided into the regulator's own printed generation it implies net fleet efficiencies inside a physically plausible band in nine of nine panel years where the aggregate conversion manages two. That conversion is retained as *grid_nepra_reference_legacy* 57.82 Mt rather than deleted. Beside them the family still carries *grid_plant_cv_restated* 38.45 Mt, the same mass at heat-rate-implied calorific values and a fuel-property alternative, and *grid_plus_ke* 44.05 Mt, a coverage alternative kept apart from it, each flagged provisional-structural-boundary in the scenario table—and each is now a precise question for the statistical authority: whether Table 3's coal row shares the plant tables' scope, and what closes the gap with the UNFCCC filing that agrees with it.

Coverage and accuracy need different criteria. Captive gas self-generation contributes 0.03 % of national inventory variance, so no variance-based ranking would ever have recommended building it. Absent categories contribute nothing to variance precisely by being absent; the completeness table, not the variance ranking, is what identifies work of this kind.

A bound labelled as a bound beats an estimate resting on an invented parameter. The air-pollutant layer is published uncontrolled—208.7 kt SO_2, 113.4 kt NO_x, 362.3 kt filterable particulate, 34 kt $PM_{2.5}$ from each coal rank's own AP-42 size distribution—because no dataset records which controls are fitted where, and an assumed removal efficiency would be indistinguishable in print from a measured one. The control-device inventory is named as what would turn the bound into an estimate.

For Pakistan the priorities are concrete, and the ledger has made them precise. Reconcile the coal scope between Table 3 and the plant tables—the heat-rate channel has already supplied the calorific values, and what remains is whether the two coal figures describe the same coal, a question worth 17.78 Mt. Establish the national fuel table's K-Electric scope, which decides whether Karachi's generation—a trending 7.1 Mt to 4.0 Mt across FY2019–24, now published as the grid-plus-K-Electric coverage alternative—is outside the activity basis. Request the licensee environmental submissions from NEPRA's HSE department—an audit of its published HSE reports shows the control-device data exists there unpublished. And extend the captive method—find the regulator who regulates the fuel rather than the plant—to industrial coal, petroleum coke and bagasse, which with process gas and the metered liquids already estimated would close most of the roughly 16 Mt still missing from Pakistani industrial combustion.

Code and data availability

The calculation engine, the activity data, the captive plant register and every output quoted here are version-controlled in the inventory repository (`https://github.com/RehanAhmad13/pakistan-emissions-sectors`) and reproduce from a single command. The sector is regenerated with `python3 -m pkinv.sectors.power.run_power`; Appendix A gives the full sequence. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

References

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US Environmental Protection Agency, 1995. AP-42: Compilation of Air Emissions Factors, Fifth Edition, Volume I, Chapter 1: External Combustion Sources. Technical Report. Office of Air Quality Planning and Standards, Research Triangle Park, NC. URL: https://www.epa.gov/air-emissions-factors-and-quantification/ap-42-compilation-air-emissions-factors. sections 1.1 bituminous and subbituminous coal, 1.3 fuel oil and 1.4 natural gas. Individual sections carry their own revision dates and are revised independently of the edition; the sections used here should be version-locked before final submission.

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Appendix A. Reproduction

The sector runs on Python 3.11 with no compiled dependencies:

```
pip install -e .
python3 -m pkinv.sectors.power.run_power
python3 -m pkinv.sectors.power.air_pollutants
```

which writes power_totals_fy2023_24.csv, power_by_stream_fy2023_24.csv, power_fuel_fy2023_24.csv, power_provincial_fy2023_24.csv, power_timeseries_fy2023_24.csv and power_air_pollutants_fy2023_24.csv to outputs/.

The run also writes the evidence tables that Section 6 and Section 7 draw on. They were not listed here before, which left every figure this paper computes from one of them without a file a reader could open:

- power_energy_closure_fy2023_24.csv: the energy closure, each item with the evidence it rests on.
- power_fuel_fy2023_24.csv: fuel by stream, and power_coal_blend_fy2023_24.csv the blend within coal.
- power_gas_technology_fy2023_24.csv: the gas fleet by technology bucket, with each bucket's energy, share and plant count.
- power_bagasse_fy2023_24.csv: the bagasse sub-ledger.
- power_heat_rate_fy2023_24.csv: the heat-rate series by fiscal year and fuel—generation fuels, TOE, net GWh, implied heat rate and efficiency—with a verdict on each row.
- power_heat_rate_stability_fy2023_24.csv: whether each fuel's efficiency is stable across the material years, as a low–high spread.
- power_hr_circularity_fy2023_24.csv: the circularity check: the ledger's own Btu/kWh against each plant's reported gross and net heat rates, so a heat rate derived from the ledger cannot be offered as evidence for it.
- power_boundary_audit_fy2023_24.csv: the boundary audit—gas and furnace oil with and without Karachi Electric, and K-E's own energy—and power_ke_leg_fy2023_24.csv that utility's leg by year with its implied net efficiency.
- power_lmdi_fy2023_24.csv: the decomposition of each year-on-year change into activity, mix and factor terms, with the coal, furnace-oil and gas mix contributions separated.
- power_point_sources_fy2023_24.csv: the point-source register: plant, fuel and energy with the town and province, and how each was located.
- power_climatetrace_plants_fy2023_24.csv: the Climate TRACE plant crosswalk, that inventory's asset against this one's ledger plant with the ratio and any flag raised.
- power_tropomi_crosscheck.csv: the satellite cross-check: observed NO_2 and SO_2 enhancement against the ledger's net generation, with the days and months behind each.
- power_monthly_fy2023_24.csv: the twelve-month layer—category, subitem and gas by month with each month's share, the emissions it implies in both common-GWP vintages, and the temporal method and basis behind the shape.
- power_uncertainty_fy2023_24.csv: the sector interval by category—the AR5 total with its Approach-1 percentage computed both ways, grouped and independent, so the correlation assumption is visible rather than chosen silently.

Figures are regenerated with python3 -m pkinv.report.figures and copied into this manuscript by python3 scripts/sync_figures.py; a test fails if any shipped figure differs from the output it plots. The full suite runs with python3 -m unittest discover tests.

Appendix B. Sources

Grid fuel is the NEPRA State of Industry Report, Table 3. Captive gas is the OGRA State of the Regulated Petroleum Industry annual reports, FY2016–24. The captive plant register combines the SNGPL captive fleet, the SSGC geocoded captive list and the NEPRA generation-CPP licence register, deduplicated; plant-wise generation for the provincial proxy comes from the same consolidated workbook. All are archived under `sources/Power Plants/` and described in `docs/sources.md`.

Emission factors are IPCC 2006 Guidelines Volume 2 Chapter 2 defaults (IPCC, 2006): Table 2.2 for grid generation, Table 2.3 for captive. Air-pollutant factors are US EPA AP-42 Sections 1.1 (bituminous coal), 1.3 (fuel oil), 1.4 (natural gas), 1.7 (lignite—79% of the fleet's coal energy since the rank split), 1.8 (bagasse) and 3.1 (stationary gas turbines), transcribed with their table numbers and quality ratings; the fetched section PDFs are hashed in `sources/AP42/`. External comparison is EDGAR (Crippa et al., 2025).

Appendix C. Supplementary Note: replacement of the provincial allocation proxy

Earlier revisions allocated grid emissions to provinces by installed capacity from a global power-plant database. That proxy predated most of the coal build-out (21.5 GW listed against roughly 45 installed) and assigned provinces by plant coordinates, which substantially overallocated to Khyber Pakhtunkhwa—a province whose electricity is almost entirely hydro, which burns nothing and belongs nowhere in a combustion sector.

The first replacement used contemporaneous plant-wise thermal generation: Punjab 49.67 %, Sindh 34.22 %, Balochistan 16.09 %, Khyber Pakhtunkhwa 0.02 % on FY2022 output, against the capacity proxy's 32.95, 43.61, 9.20 and 14.24 %.

That share was still one geography for every fuel. The current allocation is per fuel, from the FY2023–24 plant fuel ledger joined to the report's own plant-location table (Table 5), energy-weighted with each row's evidence class carried: coal Punjab 9.7 / Sindh 89.4 / Balochistan 0.8 %; gas and RLNG 59.8 / 22.8 / 17.4 %; furnace oil and diesel wholly Punjab in the year's data (Hub reports zero fuel). K-Electric plants are excluded from the shares; whether the national fuel rows already contain their fuel is not established (Section 6.9), and the coverage alternative prices it beside the reference. Bagasse rides its own all-Punjab row. The aggregate FY2022 share remains in the parameter file as the documented fallback, and both replaced proxies' provenance and caveats are preserved in its history. Each committed per-fuel share is recomputed from the ledger by a test and fails on drift.